CF Sample Imp.

In this document, we will delve into the comprehensive Implementation Steps of CF with Redirection Form and provide sample requests, responses, and best practices for a fast, easy, and secure integration.

Pre-Requirements;

Before proceeding, please ensure that you have the following pre-requisites in place:

- A sandbox account
- Pair of API & Secret keys
- A software tool to manage API calls; Postman, Insonmina, Github clients

If so, keep going.

Overview

The implementation of CF requires two subsequent `POST` requests to be orchestrated one after the other.

In short;

- First, we will illustrate a **CF-Initialize** `POST` request.
- Then, we will cover up with **CF-Retrieve** `POST` section.

Steps

1. CF-Initialize
2. Token + URL
3. Redirection
4. CF-Retrieve
5. Webhook

Step 1 - CF Initialize

CF implementation begins with PWI-Initialize `POST` request.

This step involves sending a POST request and the request body contains various parameters such as locale, conversationId, price, basketId, paymentGroup, callbackUrl, currency, paidPrice, enabledInstallments, buyer details, shipping address, billing address, and basket items.

The following object is a sample CF-**Initialize** request body.

Sample CF-Initialize Request;

```
{
   "locale":"en",
   "conversationId":"sampleConversationId",
   "price":"5.2",
   "basketId":"B67832",
   "paymentGroup":"PRODUCT",
   "buyer":{
      "id":"BY789",
      "name":"Higher",
      "surname":"Faster",
      "identityNumber":"74300864791",
      "email":"stronger@implementation.com",
      "gsmNumber":"+905555434332",
      "registrationAddress":"Nidakule Göztepe, Merdivenköy Mah.
      "city":"Istanbul",
      "country":"Turkey",
      "ip":"85.34.78.112"
   },
   "shippingAddress":{
      "address":"Nidakule Göztepe, Merdivenköy Mah. Bora Sok. N
      "contactName":"Jane Doe",
      "city":"Istanbul",
      "country":"Turkey"
   },
   "billingAddress":{
      "address":"Nidakule Göztepe, Merdivenköy Mah. Bora Sok. N
      "contactName":"Jane Doe",
      "city":"Istanbul",
      "country":"Turkey"
   },
   "basketItems":[
      {
         "id":"BI101",
         "price":"5.2",
         "name":"Binocular",
         "category1":"Category 1",
         "itemType":"PHYSICAL"
      }
   ],
   "callbackUrl":"YOUR_CALLBACKURL",
   "currency":"TRY",
   "paidPrice":"5.2",
   "paymentSource":"zooz",
   "enabledInstallments":[
       2,3
   ]
}
```

Once you have triggered above request, there are 3(three) scenarios you will monitor mostly.
Happy path includes a variable called, `paymentPageUrl` .

```
{
    "status": "success",
    "locale": "en",
    "systemTime": 1687112732811,
    "conversationId": "sampleConversationId",
    "token": "a5b67652-c24a-4347-b61e-6c957bf30f1b",
    "checkoutFormContent": "<script type=\"text/
javascript\">if (typeof iyziInit == 'undefined') {var
iyziInit = {currency:\"TRY\",token:\"a5b67652-c24a-4347-
b61e-6c957bf30f1b\",price:5.20,locale:\"en\",baseUrl:
\"https://sandbox-api.iyzipay.com\", merchantGatewayBaseUrl:
\"https://sandbox-merchantgw.iyzipay.com\",
registerCardEnabled:true,bkmEnabled:true,bankTransferEnabled:
true,bankTransferTimeLimit:{\"value\":5,\"type\":
\"day\"},bankTransferRedirectUrl:
\"YOUR_CALLBACKURL\",bankTransferCustomUIProps:
{},campaignEnabled:false,campaignMarketingUiDisplay:null,paym
entSourceName:
\"zooz\",plusInstallmentResponseList:null,payWithIyzicoSingle
Tab:true,payWithIyzicoOneTab:false,mixPaymentEnabled:true,cre
ditCardEnabled:true,bankTransferAccounts:[{\"iban\":
\"TR020006200035600006294884\",\"legalCompanyTitle\":\"iyzi
Ödeme ve Elektronik Para Hizmetleri A.Ş.\",\"currency\":
\"TRY\",\"bank\":\"Garanti Bankası\",\"bankId\":62,
\"bankLogoUrl\":\"https://static.iyzipay.com/checkoutform/
img/banks/62.svg\"},{\"iban\":\"TR080006400000111351092786\",
\"legalCompanyTitle\":\"iyzi Ödeme ve Elektronik Para
Hizmetleri A.Ş.\",\"currency\":\"TRY\",\"bank\":\"İş
Bankası\",\"bankId\":64,\"bankLogoUrl\":\"https://
static.iyzipay.com/checkoutform/img/banks/64.svg\"},
{\"iban\":\"TR340004600786888000080896\",
\"legalCompanyTitle\":\"iyzi Ödeme ve Elektronik Para
Hizmetleri A.Ş.\",\"currency\":\"TRY\",\"bank\":\"Akbank\",
\"bankId\":46,\"bankLogoUrl\":\"https://static.iyzipay.com/
checkoutform/img/banks/46.svg\"},{\"iban\":
\"TR910006701000000096073759\",\"legalCompanyTitle\":\"iyzi
Ödeme ve Elektronik Para Hizmetleri A.Ş.\",\"currency\":
\"TRY\",\"bank\":\"Yapı Kredi Bankası\",\"bankId\":67,
\"bankLogoUrl\":\"https://static.iyzipay.com/checkoutform/
img/banks/67.svg\"},{\"iban\":\"TR760001001187768823545001\",
\"legalCompanyTitle\":\"iyzi Ödeme ve Elektronik Para
Hizmetleri A.Ş.\",\"currency\":\"TRY\",\"bank\":\"Ziraat
Bankası\",\"bankId\":10,\"bankLogoUrl\":\"https://
static.iyzipay.com/checkoutform/img/
banks/10.svg\"}],userCards:
[],fundEnabled:true,memberCheckoutOtpData:
{},force3Ds:false,isSandbox:true,storeNewCardEnabled:true,pay
```

```
{},isSandbox:true,isSandboxYtb:true,threeDsForceEnabled:true,payWithNewCardEnabled:true,enabledApmTypes:[\"SOFORT\",\"IDEAL\",\"QIWI\",\"GIROPAY\"],payWithIyzicoUsed:false,payWithIyzicoEnabled:true,payWithIyzicoCustomUI:{},buyerName:\"Higher\",buyerSurname:\"Faster\",merchantInfo:\"\",merchantName:\"Sandbox Merchant Name - 3389550\",cancelUrl:\"\",buyerProtectionEnabled:false,hide3DS:false,gsmNumber:\"+905555434332\",email:\"stronger@implementation.com\",checkConsumerDetail:{},subscriptionPaymentEnabled:false,ucsEnabled:false,fingerprintEnabled:false,payWithIyzicoFirstTab:false,creditEnabled:true,payWithIyzicoLead:false,goBackUrl:\"\",metadata :{},createTag:function(){var iyziJSTag = document.createElement('script');iyziJSTag.setAttribute('src','https://sandbox-static.iyzipay.com/checkoutform/v2/bundle.js?v=1687112732810');document.head.appendChild(iyziJSTag);}};iyziInit.createTag();}</script>",
    "tokenExpireTime": 1800,
    "paymentPageUrl": "https://sandbox-cpp.iyzipay.com?token=a5b67652-c24a-4347-b61e-6c957bf30f1b&lang=en",
    "payWithIyzicoPageUrl": "https://sandbox-ode.iyzico.com/?token=a5b67652-c24a-4347-b61e-6c957bf30f1b&lang=en"
}
```

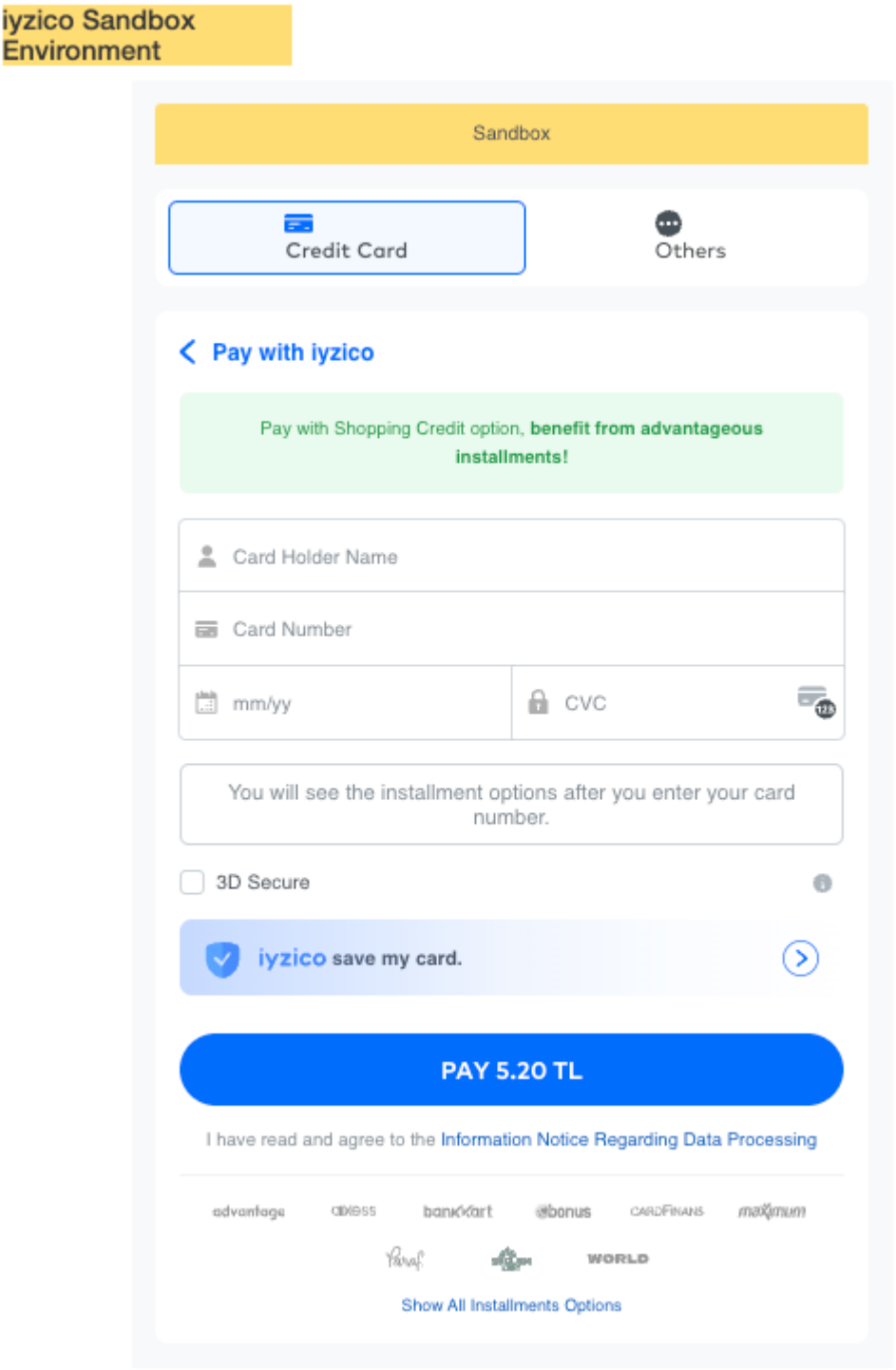

Step 2 - Token + URL

Hereby, `token` and `paymentPageUrl` are essential knots of for the rest of payment journey.

In this step, you need to extract the token and `paymentPageUrl` from the response obtained in the previous step. The token should be saved for later validation, and the `paymentPageUrl` is used to redirect the user to the payment page.

- **token;** Its expected that merchants saves `token` parameter to confront and validate payment at the very end of payment journey.
- `paymentPageUrl` **;** where payment page appears briefly. Redirection to `paymentPageUrl` provides an interface to accomplish checkouts safely for end-users.

An example of `paymentPageUrl` interface on the browser;



Step 3 - Redirection

Redirection phase ties CF-Initialize and CF-Retrieve steps overall.

Following up end user interact on CF payment page, iyzico will automatically redirect the page to the address given in the "callbackUrl" parameter.
Redirection `POST` includes;

Input Name	Type	Description
token	String	Unique value created for each checkout form request. This token is used to access the payment result

Step 4 - CF Retrieve

CF-Retrieve is the last step of PWI chain.

Basically, we have just accepted a payment on the previous step.
It is now time to legalize the transaction in this step.
The following example is a sample **CF-Retrieve** request body.

Sample CF-Retrieve Request;

```
{
    "locale": "en",
    "token": "a5b67652-c24a-4347-b61e-6c957bf30f1b",
    "conversationId": "sampleConversationId"
}
```

Once you have triggered above request, there are 2(two) scenarios you will monitor mostly.
Happy path includes `SUCCESS` parameter in the `paymentStatus` variable.

Sample CF-Retrieve Response;

```
{
    "status": "success",
    "locale": "en",
    "systemTime": 1687113618054,
    "conversationId": "sampleConversationId",
    "price": 5.20000000,
    "paidPrice": 5.20000000,
    "installment": 1,
    "paymentId": "19831123",
    "fraudStatus": 1,
    "merchantCommissionRate": 0E-8,
    "merchantCommissionRateAmount": 0E-8,
    "iyziCommissionRateAmount": 0.20800000,
    "iyziCommissionFee": 0.25000000,
    "cardType": "CREDIT_CARD",
    "cardAssociation": "TROY",
    "cardFamily": "Cardfinans",
    "binNumber": "979203",
    "lastFourDigits": "0000",
    "basketId": "B67832",
    "currency": "TRY",
    "itemTransactions": [
        {
            "itemId": "BI101",
            "paymentTransactionId": "21097896",
            "transactionStatus": 2,
            "price": 5.20000000,
            "paidPrice": 5.20000000,
            "merchantCommissionRate": 0E-8,
            "merchantCommissionRateAmount": 0E-8,
            "iyziCommissionRateAmount": 0.20800000,
            "iyziCommissionFee": 0.25000000,
            "blockageRate": 0E-8,
            "blockageRateAmountMerchant": 0E-8,
            "blockageRateAmountSubMerchant": 0E-8,
            "blockageResolvedDate": "2023-06-26 00:00:00",
            "subMerchantPrice": 0E-8,
            "subMerchantPayoutRate": 0E-8,
            "subMerchantPayoutAmount": 0E-8,
            "merchantPayoutAmount": 4.74200000,
            "convertedPayout": {
                "paidPrice": 5.20000000,
                "iyziCommissionRateAmount": 0.20800000,
                "iyziCommissionFee": 0.25000000,
                "blockageRateAmountMerchant": 0E-8,
                "blockageRateAmountSubMerchant": 0E-8,
                "subMerchantPayoutAmount": 0E-8,
```

```
                "merchantPayoutAmount": 4.74200000,
                "iyziConversionRate": 0E-8,
                "iyziConversionRateAmount": 0E-8,
                "currency": "TRY"
            }
        }
    ],
    "authCode": "104530",
    "phase": "AUTH",
    "mdStatus": 1,
    "hostReference": "mock00037iyzihostrfn",
    "token": "3ee2d355-da4d-47ce-bd3d-13d8c1854128",
    "callbackUrl": "YOUR_CALLBACKURL",
    "paymentStatus": "SUCCESS"
}
```

`mdStatus` parameter will only appear on 3DS operations.

Please check further details around mdStatus here.

Step 5 - Webhook

Webhooks are tail subject of our implementation streamline.

Receiving real-time payment notifications leverages overall CF experience while ensuring that the notifications are coming from trusted sources, preventing any potential tampering or unauthorized access to sensitive data.

Each payment event triggers webhooks;

- In `15` seconds.
- Until your server responds with `200`.
- For every `10` minutes.
- Max `3` times.

Sample PWI Webhook

Parametere	Type	Description
iyziEventTime	long	Unix timestamp value of first notification.
iyziEventType	string	Shows the request type. Values: API_AUTH, THREE_DS_AUTH, BKM_AUTH
iyziReferenceCode	string	A unique reference code for the notification
paymentId	long	Unique iyzico reference code of related payment
paymentConversationId	string	Merchant's reference id for the related payment
status	string	Payment status. Values: SUCCESS, FAILURE

Happy path includes `success` parameter in the `status` variable.

Confirmation

To finalize payment operations successfully, correlation of certain variables plays significant role;

- `paymentId`
- `status`
- `price`
- Other Idempotency parameters optionally

CF-Retrieve response and Webhook notification have to authenticate each other.

It is crucial to cross-verify the response received from CF-Retrieve and the data received in the Webhook notification. Match the paymentId, status, price, and other relevant parameters to ensure the authenticity and correctness of the transaction.

- `paymentId` on CF-Retrieve response should be equal to `paymentId` from Webhook
- Both of the `status` parameters should be `success`
- `price` parameter amounts should be the same
- And lastly, Idempotency parameters should be the same as well.

If so, congratulations! Just finished PWI implementation.

Transaction details could get from merchant panel.

> Please note that the provided sample requests and responses are for illustration purposes only, and you should adapt them to your specific implementation requirements and API documentation.

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CF-Retrieve

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Last updated 6 months ago